

People who say ai consumes gallons of water forget about the water cycle

Posted by u/shank_8 • 0 points • 19 comments

Water is used for cooling unless you don't know and when water is heated by the hardware parts it creates steam and in turn into clouds, where it rains and the cycle starts again.

Seriously i was thought about this ever since 1st grade and water is literally renewable because unless the cooling process uses electrolysis and destroys the H2O into hydrogen and oxygen then it would just go back into the atmosphere and into somewhere else. And unless the place where you get the water from is ridiculously hot and almost has no rainfall then it's basically infinite cooling as long as you could pay the bills.

Nuclear power plants use water to cool their coolants and the smoke you see isn't scary radioactive mist its steam thats the same thing happening here.

Comments

u/[deleted] • 1 points • 2026-03-09 19:20 UTC

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u/chubbipuppil • 1 points • 2026-01-23 21:06 UTC

ai and art in the same sentence 🤖 . Y'all need help. You guys aren't artists.

u/[deleted] • 1 points • 2025-10-26 06:34 UTC

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u/Complex_Captain_5923 • 1 points • 2025-07-20 22:16 UTC

Then you have people wasting water before Chatgpt

u/Helldiver_of_Mars • 2 points • 2025-04-22 04:33 UTC

Response from an AI:

Yes, AI's water usage is becoming a significant environmental issue, primarily due to the large amounts required for cooling data centers. These centers, which house powerful servers running AI models, consume extensive water resources to manage heat generated by computing hardware.

Why Water Use is an Issue:

1. Cooling Data Centers:

Large-scale AI models, like ChatGPT or GPT-4, require extensive computing power, generating substantial heat.

Data centers typically use water-intensive cooling systems to prevent overheating.

2. Scale and Growth:

Rapid growth in AI adoption has led to increased numbers and sizes of data centers globally.

The water use from a single large data center can be substantial—some reports suggest millions of gallons per month.

3. Regional Stress:

Many data centers are located in regions already facing water scarcity, exacerbating local environmental challenges.

Communities in arid regions have expressed concern about sustainability and equitable access to local water resources.

4. Indirect Impact:

Beyond direct cooling, the generation of electricity powering data centers often requires significant water, especially when relying on coal, natural gas, or nuclear power plants.

Current Response and Mitigation:

Improved Cooling Technology:

Companies are exploring alternatives such as air-cooled systems, liquid immersion cooling, or closed-loop cooling systems that minimize water loss.

Data Center Placement:

Increasingly, companies seek locations with ample renewable energy and lower water stress.

Transparency and Regulation:

Public pressure and regulatory initiatives push technology companies toward transparency about resource consumption and incentivize reduced water usage.

In short, AI's water usage is a valid environmental concern. Addressing it involves advancing cooling technology, strategic placement of infrastructure, and greater transparency from AI companies.

u/[deleted] • 1 points • 2026-02-12 18:04 UTC

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u/Leo-Len • 2 points • 2025-04-22 03:52 UTC

This is not the issue. The issue is both the buying up of water supplies in local communities and the heat pollution in open water systems. The latter is an issue in lakes housing an ecosystem as it can completely change the local environment. Not good.

Also, energy is a bigger concern than water by a large factor. A single chatgpt query for instance takes up around 0.03 watt hours (<https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use>). This isn't a lot on its own, but when you consider the fact that search engines are beginning to affect built in AI to their queries, it starts to stack up.

u/No_Damage9784 • 2 points • 2025-04-21 20:14 UTC

I'm not gonna lie I forgot about the water cycle lol but I agree with you and also yea their water argument is very ignorant

u/damontoo • 2 points • 2025-04-21 16:25 UTC

It's estimated that 5-50 ChatGPT prompts consume 500ml to process. A single quarter pound hamburger patty consumes 500 gallons to produce. Using the worst estimate of 5 prompts, one hamburger is roughly equivalent to 19,000 ChatGPT prompts.